

Lab 2: Real Time Kinematic GNSS

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Introduction.

Real Time Kinematic (RTK) is a specialized form of differential GPS (DGPS), a method for improving accuracy that involves a second reference station whose position is accurately known. By calculating a correction factor based on pseudo ranges from each satellite, DGPS enables the user station to apply those correction factors and generate position estimates with higher accuracy than standard GNSS. Specifically, RTK uses carrier phase measurements to improve the base station's accuracy and can provide accuracy to the centimeter level if the fix mode is initialized correctly.

Figure 1. *Scatter Plots of Occluded and Open Stationary Datasets with RTK.*

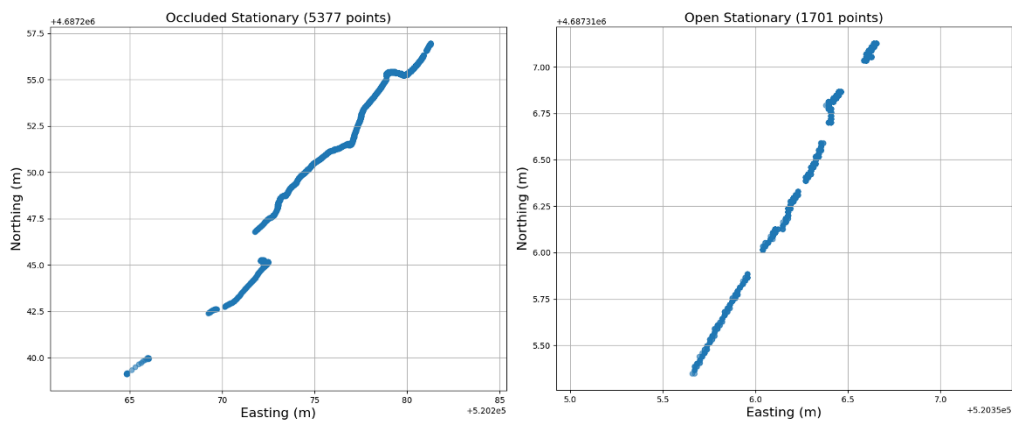
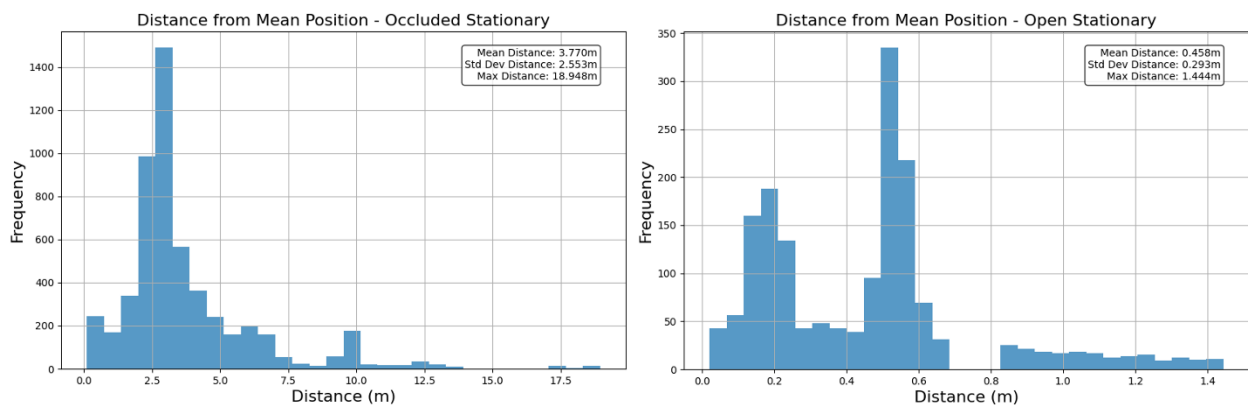


Figure 2. *Histograms of Occluded and Open Stationary Datasets with RTK.*



Discussion: Parts A, B, C, E.

The main difference between observed error in the data collected with RTK versus standard GNSS is the order of magnitude of the values. As advertised by [1], the mean RMS errors for walking trials were all below one meter. While the error statistics for the stationary trials were comparable in the occluded area for RTK and standard GNSS, the RTK open trials revealed an increase in accuracy up to 10^3 in magnitude (Table 1). Figure 2 visualizes the distribution of Euclidean distances from the mean

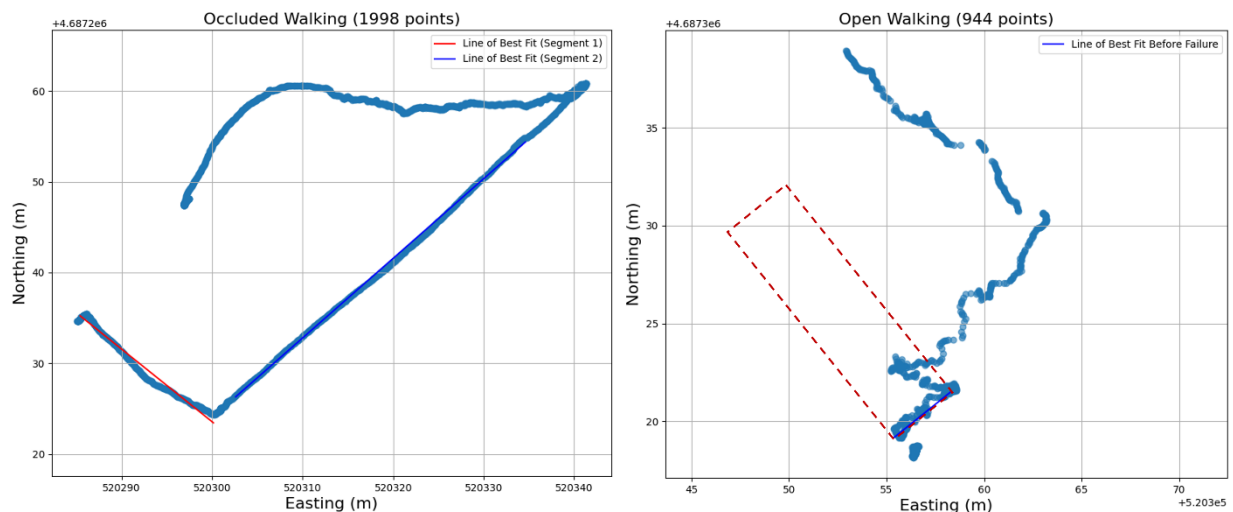
position of the data, revealing unimodal (occluded, left) and bimodal (open, right) behavior, whereas Lab 1 histograms were multimodal. This may suggest that the RTK system is better equipped to mitigate sources of noise, as the number of modes of a normal distribution can represent the number of underlying processes that influence the dataset. Given that occlusions may cause a reduction in satellite visibility, it is reasonable to attribute some of the error in occluded stationary data to drops in RTK fixes, preventing correction factors from being applied reliably. The dataset was collected between ISEC and EXP on a cold day, and after verifying that a fix was obtained at the start of recording, the device was monitored from a window indoors where the fix status was not visible. Additionally, both RTK and standard GNSS utilize Kalman filtering, a linearization method that does not accurately model stationary data and will also contribute to the observed error. Since we cannot verify that correction errors were applied sufficiently enough to improve accuracy, similarities across GNSS and RTK stationary data may be expected.

Table 1. Stationary Dataset: Distance from Mean Position Across Lab 1 (GNSS) & Lab 2 (RTK).

	Occluded		Open	
	RTK	GNSS	RTK	GNSS
Mean (m)	3.77	5.46	0.458	79.8
Standard deviation (m)	2.55	0.220*	0.293	0.35*
Maximum (m)	19.0	5.78	1.44	80.2

*Note. Standard deviation reported in Lab 1 was computed against the mean of the dataset error to reflect the precision of the measurements, not deviation from the known position.

Figure 4. Scatter Plots of Occluded and Open Walking Datasets with RTK.



d. Give quantitative comparisons for how your moving data differ in the open and occluded cases, including error/deviation estimates? Does this have anything to do with GNSS fix quality?

Figure 4 shows the RTK trajectories during our walking trials. Our occluded trial (left) was recorded between ISEC and EXP. The trajectory began as a rectangle; however, due to pedestrian traffic and the geometry of the traversable area, we deviated from the planned path and recorded the second half on a nonlinear trajectory to return to the initial position. From observation, the recorded trajectory appears visually accurate to the actual trajectory until the data stream drops off before the loop was closed, in an area of the path closest to ISEC where occlusion error is most likely to impact visibility and estimation quality. Since linear regression would not provide valuable information for the nonlinear portion, lines of best fit and RMS values were computed for the first half of the dataset only. The rectangular portion was modeled and analyzed as a function of two linear piecewise equations. The RMS values for the first (width of the rectangle) and second (length) segments were 0.521 and 0.284 meters, respectively.

Our open trial (right) appears to have lost its fix after during the second segment of the rectangle, and then re-initialized afterwards. The dotted red path represents the approximate known trajectory. Splitting the dataset to analyze it as a piecewise function was difficult as we were forced to arbitrarily shorten the time intervals due to space constraints, so linear regression analysis was performed on the first segment, where the fix mode was verifiably initialized and the time interval was known. The RMS value for the first segment was 0.452 meters, aligning with the expected accuracy level with RTK [1].

While the RMS values across the valid portions of the occluded and open trajectories were similar, the large gap in the data stream of the occluded dataset speaks to how occlusions may reduce the quality of position readings. The second leg of the occluded trial was recorded on the sidewalk along Columbus Avenue, where occlusions may not pose as large an impact as it would the location of the data stream gap.

References

- [1] van Diggelen, Frank, "GPS and GPS+GLONASS RTK," *Proceedings of the 10th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GPS 1997)*, Kansas City, MO, September 1997, pp. 139-144.